87 Prince St, Mullumbimby: deck and awning footings

For **Old Mill, Mullumbimby** · Bryce Noakes · 2 Nov 2026

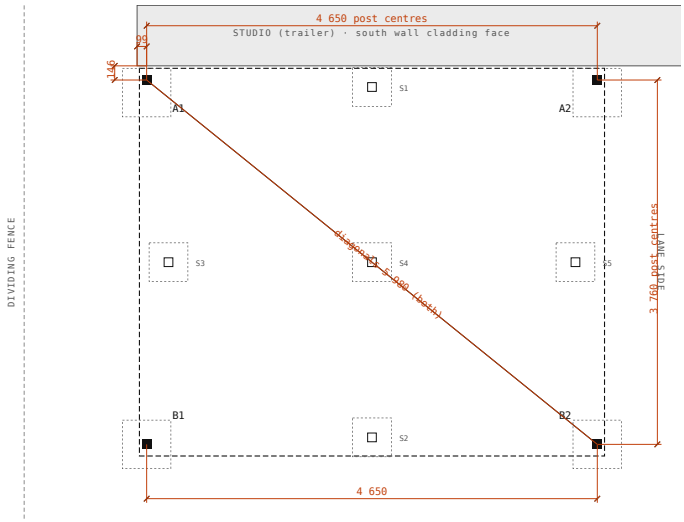
Design reviewed by **Ollie Harris, structural engineer** (markup and phone call, 2 Nov 2026). Everything below follows his notes; anything he didn't flag is as drawn and approved.

3D model, plans and engineering

TODAY: CORNER POSTS AND FOOTINGS (ENGINEER'S METHOD, 2 NOV CALL)

Item	Spec (engineer)	Qty	Old Mill: have it / sub
Corner posts	**100 × 100 hardwood, in-ground durability class** (cast straight into concrete, no brackets). Engineer: a cast-in post must be H4 or an in-ground class hardwood.	2 × 4.2 m 2 × 3.9 m	
Concrete, corner pads	20 kg bags. Four pads 500 × 500 × 250 = 0.25 m³	30	
Concrete, stump pads	20 kg bags. Five pads 400 × 400 × 200 = 0.16 m³	20	
Stump anchors	Short galvanised post anchor for 90 × 90 stumps, M12, to fit a 200 deep pad with concrete under the stem	5	
Stumps	90 × 90 F7 H4 × 2.7 m (makes all five)	1	
Bitumen paint	For the buried post ends (already bought)	—	

FOOTING SETOUT



POST CENTRES (MM)

Post	Along studio wall from its west end	Out from studio cladding
A1	99	146
A2	4 749	146
B1	99	3 906
B2	4 749	3 906

Check: 4 650 along, 3 760 across, both diagonals 5 980.

CORNER FOOTING (ENGINEER)

1. Dig **500 × 500, 800 deep** into the clay.
2. Bitumen the buried part of the post.
3. Hang the post plumb with its foot **150 off the bottom**; prop both ways.
4. Pour a **250 concrete pad**: 150 under the post, post 100 in.
5. Once set, backfill **≈550 of soil** over the pad, rammed in layers. The soil gives the uplift hold-down.

Stumps: 400 × 400 pads, 200 deep, with the anchor cast in.

Studio-side holes A1, A2: centred on the post, a 500 hole would run about 100 under the studio. Dig them from the studio line outward so the post sits off-centre in the pad, and keep clear of the trailer stands.

Full material list for quote: deck and awning

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TIMBER

Item	Spec	Length	Qty	Old Mill price / sub
Posts	100 × 100 hardwood, in-ground durability class, cast in	4.2 / 3.9 m	2 + 2	
Roof beams B1 and B2	**240 × 45 H3 MGP12** (2 plies per beam, one each side of the posts)	6.0 m	4	
Rafters	190 × 45 H3 MGP10, 6 at 930 centres	6.0 m	6	
Fascia	140 × 35 H3	6.0 m	1	
Purlins	70 × 35 H3	6.0 m	8	
Deck bearers	190 × 45 H3 MGP10, continuous: **single** on lines A and B, **double** on line C	4.8 m	4	
Deck joists	140 × 45 H3, continuous over the span, max 450 centres (engineer: keep the bigger joists for the laundry and kitchen)	4.8 m	13	
Knee braces	Min 120 × 45 H3, at all 4 corner posts, M12 bolt each end	4.8 m	1	
Stumps	90 × 90 F7 H4 (makes all five)	2.7 m	1	
Decking	90 × 22 H3 pine	5.4 m	42	
Screen battens	70 × 45 H3, on edge (vertical screen, rails, hangers)	4.8 m	25	

HARDWARE AND CONCRETE

Item	Spec	Qty	Old Mill price / sub
Stump anchors	Short galvanised post anchor for 90 stumps, M12, to suit a 200 deep pad	5	
Bolts	M12 HDG: beams to posts, bearers to posts and stumps, 8 for the knee braces; 50 × 50 square washers	1 lot	
Straps and screws	Triple grips or cyclone straps (every rafter to the beams), batten and decking screws	1 lot	
Concrete	20 kg bags: 30 for the corner pads, 20 for the stump pads	50	
End sealer	For cut ends of treated timber	1	

Roof sheets (Zincalume and clear corrugated) are not in this list.

ENGINEER'S CHANGES (OLLIE HARRIS, 2 NOV 2026)

- Bearer lines A and B: **single 190 × 45 MGP10, continuous**, inside face of the posts.
- Bearer line C: **double 190 × 45 MGP10, continuous**, on 3 stumps.
- All bearers continuous and supported by an intermediate post.
- Joists continuous over the span, max 450 centres; keep the bigger size.
- **Diagonal timber knee brace** at all 4 corner posts: min 120 × 45, M12 bolt each end.
- Posts **100 × 100 in-ground hardwood** cast straight in: 800 hole, 250 pad (150 under the post), ≈550 soil backfill. Stump pads 400 square × 200.
- B1 and B2: **2/240 × 45 MGP12**.
- Rafters 190 × 45 at 930 and fascia 140 × 35: approved as drawn.

For Old Mill: if you can't get 240 × 45 MGP12 H3 or in-ground class 100 × 100 hardwood posts, note the substitute in the right-hand column and I'll check it with the engineer before ordering. Plans, 3D model and the engineer's markup: **tradearts.work/prince-street-deck-awning/p26.html**